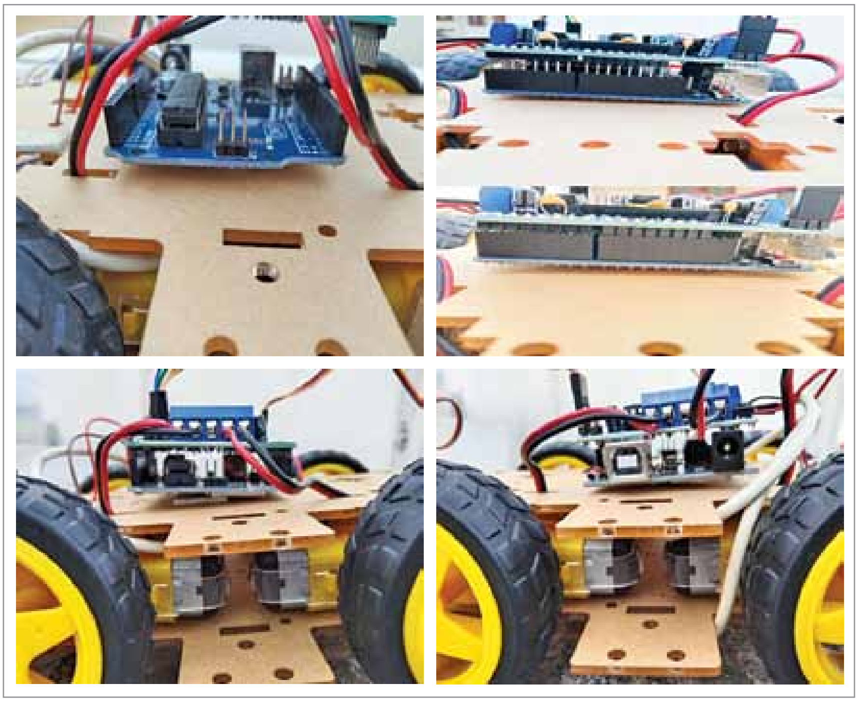
Fig. 4: Soldering wire to a gear motor

Fig. 5: Attaching gear motor to chassis

Fig. 6: Attaching wheel to gear motor



Figs 7, 8, and 9: Chassis assembly as per Step 2

avoiding collisions. Obstacle avoidance in robots can bring more flexibility in varying environments and make them much more efficient, so that their continuous human monitoring is not required. The authors' prototype is show in Fig. 1.

Circuit and working

Block diagram of the obstacle avoiding robot is shown in Fig. 2 while its circuit diagram is shown in Fig. 3. This robot is constructed using ATmega 8 family's microcontroller Arduino Uno R3. When its ultrasonic sensor detects an obstacle or an edge, it sends a command to the microcontroller. The microcontroller, based on the received input signal, directs the robot to push in an alternative direction by actuating the motors that are interfaced with it via a motor driver. Depending on the situation, the robot is able to choose a clear path and the decision-making process of obstacle avoidance and edge detection occurs spontaneously.

The sensor plays an important role in this project. First, the transmitter sends ultrasonic signal bursts at 40kHz. This signal gets reflected back when an object comes in front of the sensor and it is received by the receiver. The ultrasonic sensor generates high-frequency sound waves and evaluates the echo that reaches back to the receiver. So, by using this technique, the ultrasonic waves calculate the distance between the car and the reflected objects continuously.

After an obstacle detection, the car changes its direction by making an autonomous decision. It can measure the distance between itself and surrounding objects in real time.

Software

The program is written in Arduino programming language Sketch. Arduino IDE 1.8.11 has been used to compile and upload the program in the prototype. The project needs various libraries, which have to be included in the external header file for programming.

For uploading the source code ARDUINO_OBSTACLE_AVOIDING_CAR.ino into Arduino Uno board, connect the Arduino board to desktop/laptop using a USB cable. Next, select the board and port and upload the source code.

Construction and testing

First of all, upload the source code into the Arduino Uno and follow the steps mentioned below to assemble the robot.

Step 1. *Assembling the chassis (see Fig. 4, Fig. 5, and Fig. 6):*

• Solder the wire to positive and GND of the DC gear motors

• Attach the DC gear motors to